

Layer-Wise Gender Bias in Multilingual Foundation Models: A Five-Architecture Cross-Lingual Analysis

Seminar Paper

submitted by

Daniel Tesfai Kebede

Matr. 1716694

s4dakebe@uni-trier.de

Juan Jose Arango Serrano

Matr. 1861694

s2juaran@uni-trier.de

Hadil Athmouni

Matr. 1762515

s2haathm@uni-trier.de

Course: Trends in Machine Learning

Submission: 28.03.2026

Instructor: Raghvi Baloni

Abstract

Multilingual transformer models compress dozens of languages into shared representational spaces, enabling cross-lingual transfer while raising a structural question: when languages with radically unequal training data coexist in the same geometric space, how are social associations distributed across them? Gender bias in English embeddings has been extensively studied, yet comparatively little work examines how contextual multilingual models encode such associations in genuinely low-resource, non-Latin-script languages, or how those associations shift across model depth.

This study takes Tigrigna, a Semitic language written in the Ge'ez script and spoken primarily in Eritrea and Ethiopia, as its case study. The models evaluated here were not trained on any dedicated Tigrigna corpus, making this a genuine near-zero-shot setting where any detected bias must arise from cross-lingual transfer rather than language-specific statistics. Comparative analysis is conducted across Arabic and Spanish. We evaluate five multilingual transformer architectures without fine-tuning: `bert-base-multilingual-cased`, `xlm-roberta-base`, `xlm-roberta-large`, `facebook/xlm-v-base`, and `microsoft/mdebarta-v3-base`. Contextual token representations are reduced to stable word-level embeddings via contextual-to-static aggregation, enabling layer-wise projection of profession terms onto a geometric gender direction.

Our results have four major consequences. First, we see that, apart from one, the four models demonstrate a detectable male-leaning bias in Tigrigna profession terms, even in the absence of Tigrigna in the pretraining data, thus providing evidence of cross-lingual transfer of social bias. Second, bias builds up over layers, peaks in layer 9 or layer 23 in XLM-R-large, and then collapses in the final layer. Third, two different geometric methods agree, achieving $\rho \geq 0.964$ in all valid models and languages, thus providing evidence of the reliability of our approach. Fourth, but most interestingly, we observe a complete flip in bias in mDeBERTa-v3-base, which was trained on a very similar multilingual corpus, to a female-leaning bias, demonstrated by 28 out of 30 Tigrigna profession terms, achieving a mean projection value of -1.84 in the peak layer. This is due to the Disentangled Attention mechanism in mDeBERTa, a finding confirmed in Arabic and Spanish as well. All experiments are performed on CPU, without fine-tuning, using the publicly available checkpoints.

1 Introduction

Tigrigna, spoken by millions across Eritrea and Ethiopia, is one of the most underrepresented languages in NLP research. It has no large-scale parallel corpora, minimal presence in multilingual benchmarks, and its speakers have no guarantee that the models they interact with handle their language with the same reliability afforded to high-resource languages. What social associations do multilingual models carry into a language they were hardly trained on? This asymmetry poses a question that extends beyond performance metrics.

Prior work on gender bias in language models has focused almost exclusively on English static embeddings. The geometric bias literature establishes that social associations are recoverable from the spatial structure of representations, but it does not address whether those associations transfer across scripts, morphological types, or resource levels. Multilingual contextual models introduce further complexity: representations are built layer by layer, and different model architectures process the same input through fundamentally different attention mechanisms.

This research aims to fill the identified gap with a layer-wise, cross-linguistic, architecture-comparative study, with Tigrigna as the primary focus. The language was chosen for a specific reason: a language that is quite distant from the training set's language distribution makes one wonder what the model really knows, as opposed to what it learned from the language.

1.1 Research Questions

- **RQ1:** Do multilingual transformers encode measurable gender bias in Tigrigna, Arabic, and Spanish under a unified geometric protocol?
- **RQ2:** How does measured bias vary across transformer layers, and at which depths do associations peak?

- **RQ3:** Do two independent geometric estimators agree on bias trends across languages and models?
- **RQ4:** Does attention architecture determine gender bias direction, independent of pretraining data?

2 Related Work

2.1 Bias in Static Word Embeddings

Early work established that social biases are encoded geometrically in static word embeddings. [Bolukbasi et al. \(2016\)](#) derived a gender direction from anchor sets, revealing systematic profession–gender associations and proposing projection-based debiasing. [Caliskan et al. \(2017\)](#) introduced WEAT as a statistical framework for implicit associations; [Garg et al. \(2018\)](#) showed embedding bias correlates with historical trends. [Gonen and Goldberg \(2019\)](#) demonstrated that removing a single bias direction does not eliminate deeper geometric bias, motivating continued geometric analysis.

2.2 Interpreting Contextual Representations via Reductions

Transformer models produce contextualised representations that vary across inputs and depth, complicating direct application of geometric methods. [Bommasani et al. \(2020\)](#) formalised reductions of contextual embeddings to stable word-level vectors suitable for geometric analysis, while [Rogers et al. \(2020\)](#) surveys the uneven distribution of linguistic information across transformer layers, motivating layer-wise analysis.

2.3 Multilingual Bias and Cross-Lingual Considerations

As multilingual transformers became widely adopted, bias analysis extended beyond English. [Zhao et al. \(2019\)](#) and [Zhao et al. \(2020\)](#) confirmed that geometric bias methods transfer to contextual multilingual models and that bias patterns vary substantially across languages, with debiasing in one language not generalising to others. [Guo and Caliskan \(2021\)](#) further showed contextualised embeddings encode intersectional associations. These studies predominantly evaluate final-layer representations. [Lauscher et al. \(2020\)](#) showed zero-shot cross-lingual transfer degrades substantially for typologically distant and low-resource languages, reinforcing why Tigrigna provides a stringent near-zero-shot evaluation. Systematic layer-wise and architecture-controlled bias studies remain absent from the multilingual bias literature.

2.4 Architecture-Driven Effects on Representations

The models evaluated here differ not only in scale but in fundamental attention design. [He et al. \(2021\)](#) introduced Disentangled Attention, which separates content and positional information into independent attention matrices, substantially changing how contextual relationships are encoded across layers. [He et al. \(2023\)](#) extended this design to multilingual pretraining via an ELECTRA-style training objective, producing the microsoft/mdeberta-v3-base checkpoint evaluated here. [Liang et al. \(2023\)](#) enlarged the SentencePiece vocabulary from 250K tokens (XLM-R) to 1M tokens (XLM-V) to improve tokenisation coverage for low-resource languages. These architectural and vocabulary decisions constitute the design space whose bias consequences this study directly compares.

2.5 How This Work Differs

We integrate geometric bias analysis ([Bolukbasi et al., 2016](#)) with contextual-to-static reductions ([Bommasani et al., 2020](#)) in a controlled layer-wise, architecture-comparative study across five multilingual models, revealing that Disentangled Attention produces a complete reversal of gender bias direction not previously documented in the multilingual bias literature.

3 Methodology

3.1 Models

We evaluate five pretrained multilingual transformer models without fine-tuning. All checkpoints are loaded from HuggingFace Hub; no parameter updates are performed.

- **mBERT** (bert-base-multilingual-cased, [Devlin et al. 2019](#)): 12 transformer layers + embedding layer (13 hidden states), WordPiece vocabulary of $\approx 120K$ tokens covering 104 languages.
- **XLM-RoBERTa-base** (xlm-roberta-base, [Conneau et al. 2020](#)): 12 layers (13 hidden states), SentencePiece vocabulary of 250K tokens, trained on CommonCrawl across 100 languages.
- **XLM-RoBERTa-large** (xlm-roberta-large, [Conneau et al. 2020](#)): 24 layers (25 hidden states), same 250K SentencePiece vocabulary, larger model scale.
- **XLM-V-base** (facebook/xlm-v-base, [Liang et al. 2023](#)): 12 layers (13 hidden states), SentencePiece vocabulary of 1M tokens, designed to improve tokenisation coverage for low-resource languages.
- **mDeBERTa-v3-base** (microsoft/mdeberta-v3-base, [He et al. 2023](#)): 12 layers (13 hidden states), SentencePiece vocabulary, uses Disentangled Attention with separate content and position matrices, the key architectural difference examined in RQ4.

All experiments are implemented in Python and the full pipeline is publicly available at <https://github.com/HDGeeks/ML4NLU-PAPER>.

3.2 Languages and Data

We analyse three languages: Spanish (es), Arabic (ar), and Tigrigna (ti). Tigrigna is the primary case study: it is a Semitic language written in the Ge’ez (Ethiopic) script, with negligible dedicated presence in multilingual pretraining corpora, making it a stringent near-zero-shot evaluation.

For each language we provide a sentence-level corpus, a list of 30 profession terms, and gender anchor sets (11 male, 10 female for Tigrigna).

Since no suitable Tigrigna text corpus is publicly available, the corpus was constructed synthetically using a two-layer template approach. Layer 1 generates 2,000 profession-only sentences, placing each of the 30 profession terms in varied grammatical positions (subject, object, contextual). Layer 2 generates anchor-profession co-occurrence sentences: each of the 12 anchor pairs (male/female form) is combined with each of the 30 professions across 18 templates, producing $12 \times 30 \times 18 \times 2 = 12,960$ sentences. The total corpus is 14,960 sentences. Because Tigrigna profession terms are grammatically gender-neutral with gender expressed through verb conjugation, all templates were authored as matched male/female conjugation pairs. The templates were constructed and verified by the first author, a native Tigrigna speaker, ensuring grammatical correctness and cultural plausibility.

3.3 Context Sampling

For each target word w , we retrieve up to n sentences containing w via substring matching. We use $n = 20$ contexts per profession term and $n = 12$ contexts per anchor term across all languages, to ensure stable centroid estimation within CPU memory constraints. Words with fewer than the minimum available contexts are excluded.

3.4 Contextual-to-Static Reduction

To enable geometric analysis, we adopt the reduction framework of [Bommasani et al. \(2020\)](#), aggregating contextual embeddings to stable word-level vectors.

For each occurrence of word w in context c_i , we extract hidden states at every transformer layer. Subword token representations corresponding to w are mean-pooled across the subword fragments; the result is then

averaged across n contexts. Formally, at layer ℓ :

$$\mathbf{s}_\ell(w) = \frac{1}{n} \sum_{i=1}^n \mathbf{h}_\ell(w, c_i), \quad (1)$$

where $\mathbf{h}_\ell(w, c_i)$ is the mean-pooled subword representation of w at layer ℓ in context c_i (Bommasani et al., 2020). This yields one static embedding per layer for each target word.

3.5 Gender Direction Construction

Following Bolukbasi et al. (2016), we define a gender direction using predefined anchor sets. Let $\mathcal{M}$ and $\mathcal{F}$ denote the male and female anchor term sets. For each layer ℓ , anchor representations are reduced as in Eq. (1). Layer-specific centroids are:

$$\mathbf{c}_{M,\ell} = \frac{1}{|\mathcal{M}|} \sum_{w \in \mathcal{M}} \mathbf{s}_\ell(w), \quad \mathbf{c}_{F,\ell} = \frac{1}{|\mathcal{F}|} \sum_{w \in \mathcal{F}} \mathbf{s}_\ell(w), \quad (2)$$

and the layer-specific gender direction is:

$$\mathbf{g}_\ell = \mathbf{c}_{M,\ell} - \mathbf{c}_{F,\ell}. \quad (3)$$

This vector encodes the principal geometric contrast between male and female anchors at layer ℓ (Bolukbasi et al., 2016).

3.6 Bias Measurement

For each profession p and layer ℓ we compute two geometric estimators.

(1) Gender Direction Projection (proj). The scalar projection of $\mathbf{s}_\ell(p)$ onto the gender direction:

$$\text{ProjBias}_\ell(p) = \frac{\mathbf{s}_\ell(p) \cdot \mathbf{g}_\ell}{\|\mathbf{g}_\ell\|}. \quad (4)$$

Positive values indicate alignment toward the male centroid; negative values indicate alignment toward the female centroid. The scalar projection preserves vector magnitude and is sensitive to the strength of the gender association (Bolukbasi et al., 2016).

(2) Centroid Cosine Difference (cosdiff). As a complementary magnitude-normalised estimator:

$$\text{CosBias}_\ell(p) = \cos(\mathbf{s}_\ell(p), \mathbf{c}_{M,\ell}) - \cos(\mathbf{s}_\ell(p), \mathbf{c}_{F,\ell}), \quad (5)$$

where cosine similarity is defined as (Manning et al., 2008):

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}. \quad (6)$$

This estimator directly compares relative angular proximity to male and female centroids and is invariant to vector scale.

3.7 Estimator Agreement (RQ3)

To assess measurement robustness, we compute Spearman rank correlation between the two estimators across profession terms at each layer (Spearman, 1904):

$$\rho_\ell = \text{Spearman}\left(\left\{\text{ProjBias}_\ell(p)\right\}_p, \left\{\text{CosBias}_\ell(p)\right\}_p\right). \quad (7)$$

$\rho_\ell = 1$ indicates perfect ordinal agreement across professions.

3.8 Layer-Wise Aggregation

For visualisation we compute the mean projection score across retained profession terms at each layer ℓ , yielding a layer-wise bias trajectory:

$$\overline{\text{Proj}}_\ell = \frac{1}{|P|} \sum_{p \in P} \text{ProjBias}_\ell(p), \quad (8)$$

where P is the set of retained professions. This trajectory summarises how the overall geometric gender signal evolves across model depth.

4 Results

4.1 Data Coverage After Context Filtering

Table 1 reports vocabulary and corpus coverage after applying the minimum context requirement.

Table 1: Coverage after context filtering for all three languages.

Language	Corpus size	Professions retained	Male anchors	Female anchors	Contexts per profession
Arabic	188,414	23/24 [†]	12	11 [‡]	20
Spanish	100,000	24/30	11	11	20
Tigrigna	14,960	30/30	11	10	20

[†] إطفائي (firefighter) skipped across all models: only 2 corpus hits.

[‡] خالة (maternal aunt) excluded: only 2 corpus occurrences (threshold = 12).

Spanish: 6 professions excluded due to insufficient corpus support (fewer than 20 hits).

Under XLM-V-base, two terms, ጠበቅ (lawyer) and ነዳቋ (brick-layer), yielded zero corpus hits due to tokenisation mismatches introduced by the 1M SentencePiece vocabulary (see Section 4.7), leaving 28 terms for that model. All other models retained the full 30 terms.

4.2 Four-Model Comparison

Table 2 summarises the bias profile of each architecture across all three languages, constituting the main quantitative result.

Table 2: Gender bias profiles: four models $\times$ three languages. mBERT excluded (see Appendix A).

Model	Min ρ	Tigrigna (ti)		Arabic (ar)		Spanish (es)	
		Peak Proj	M:F:N	Peak Proj	M:F:N	Peak Proj	M:F:N
XLM-R-base	0.992	+2.99	25:5:0	+2.64	19:4:0	+2.21	16:8:0
XLM-R-large	0.993	+3.46	26:4:0	+4.54	23:0:0	+1.16	16:8:0
XLM-V-base	0.970*	+2.63†	26:2:0	+2.49*	21:2:0	+0.89*	16:9:0
mDeBERTa-v3-base	0.974	−1.84	2:28:0	−3.04*	4:19:0	−0.62‡	11:13:0

Peak Proj = mean projection at peak layer (Eq. 4). **M:F:N** = male : female : neutral professions by sign of layer-mean projection (all valid layers). **Min ρ** = minimum Spearman ρ across valid layers (Tigrigna).

* Layer 12 excluded: representational collapse (XLM-V all languages; mDeBERTa Arabic/Spanish).

† 28/30 professions; ጠበቅ and ነዳቋ excluded (0 corpus hits under 1M vocabulary).

‡ Weaker reversal: grammatically gendered profession pairs (e.g. *abogado/abogada*) dilute the layer-mean signal.

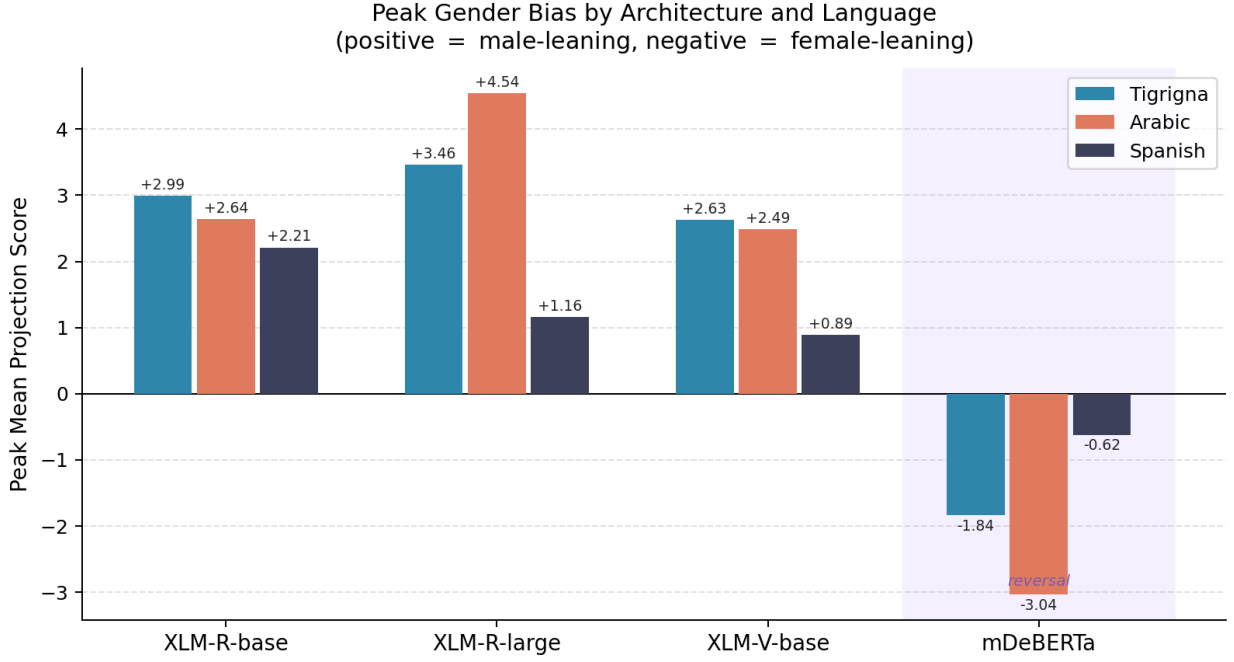


Figure 1: Peak mean projection score per architecture across all three languages. Bars below zero indicate female-leaning bias. The mDeBERTa column (shaded) shows a complete directional reversal relative to the XLM-R family, consistent across Tigrigna, Arabic, and Spanish.

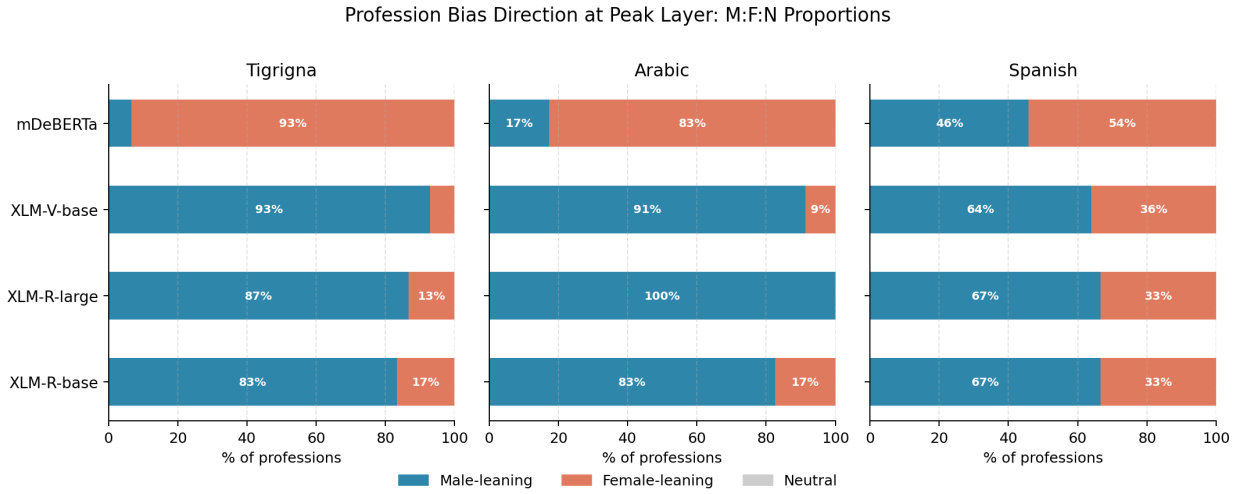


Figure 2: Proportion of professions classified as male-leaning or female-leaning across three languages. Each profession is classified by the sign of its mean projection score averaged across all valid layers (layer 12 excluded where collapse occurs). The XLM-R family assigns a male-leaning mean to 82–100% of professions in Tigrigna and Arabic; mDeBERTa inverts this pattern, with female-leaning means dominating in all three languages.

4.3 Layer-Wise Bias Trajectories

Figures 3–5 display the layer-wise mean projection trajectory for all four architectures (mBERT excluded; see Appendix A), one figure per language. Each 2×2 panel shows both base- and large-scale models for direct scale comparison.

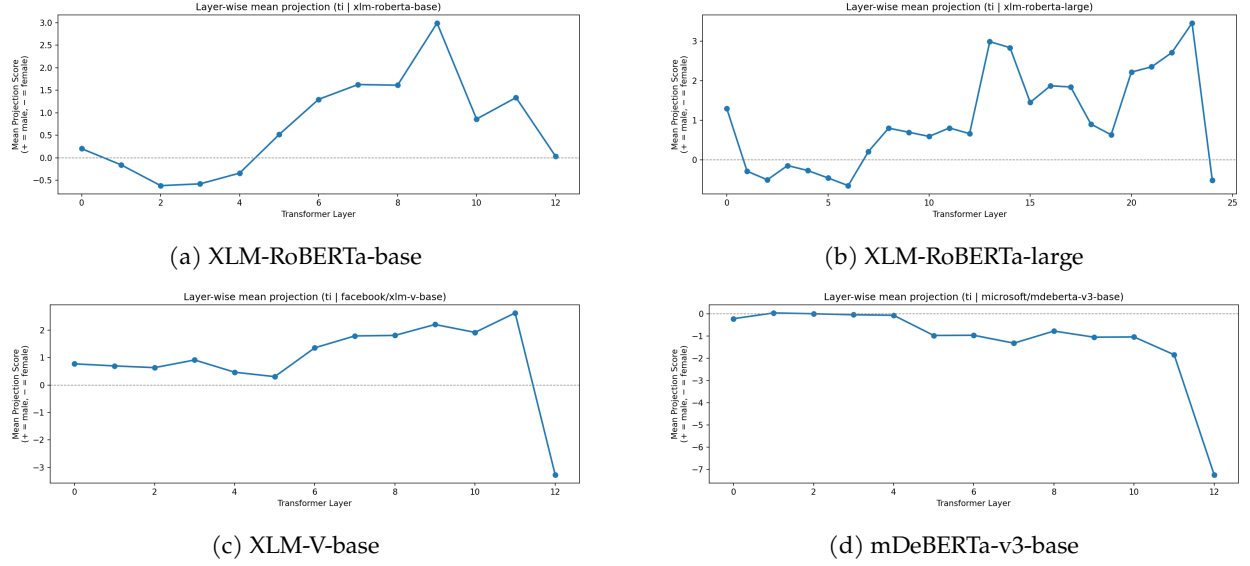


Figure 3: Layer-wise mean projection bias: **Tigrigna (ti)**. Positive = male-leaning; negative = female-leaning. Layers indexed from embedding layer (0) to final transformer block.

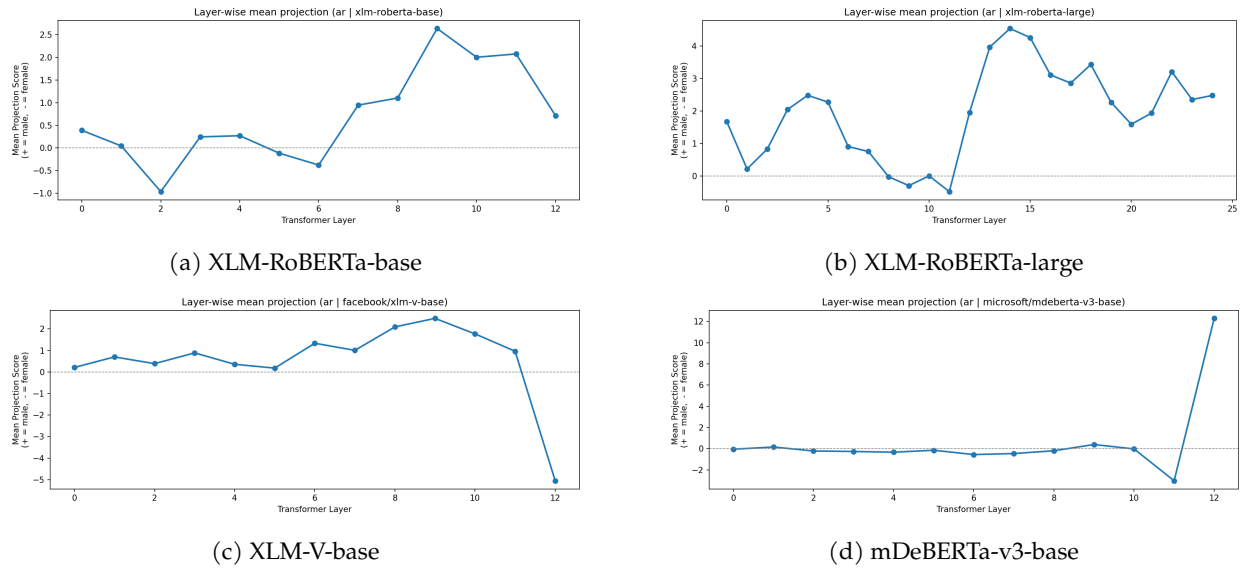


Figure 4: Layer-wise mean projection bias: **Arabic (ar)**. Positive = male-leaning; negative = female-leaning. Layers indexed from embedding layer (0) to final transformer block.

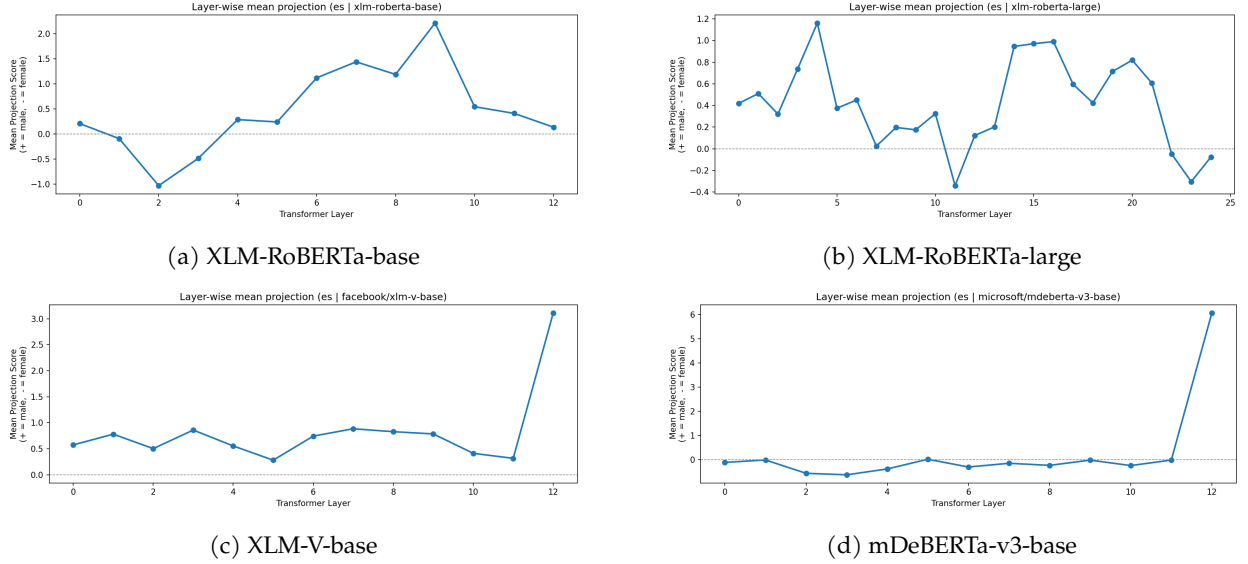


Figure 5: Layer-wise mean projection bias: **Spanish (es)**. Positive = male-leaning; negative = female-leaning. Layers indexed from embedding layer (0) to final transformer block.

Shared patterns across models. Four findings are consistent across the XLM-R family (XLM-R-base, XLM-R-large, XLM-V-base):

1. Bias is already present at layer 0 in some models and builds progressively across depth.
2. It peaks at a middle-to-upper layer (7–11 for base-size models, 14 and 23 for XLM-R-large across languages).
3. It collapses at the final layer (layer 12 for base models, layer 24 for large) toward near-zero or negative values.
4. Increasing model scale amplifies the peak magnitude ($+2.99 \rightarrow +3.46$) without changing direction.

4.4 Most Biased Professions at Peak Layer

Table 3 lists the five most male-biased professions at peak layer for XLM-RoBERTa-large, across all three languages.

Table 3: Top-5 most male-biased professions at peak layer, XLM-RoBERTa-large. ProjBias (Eq. 4). Peak layer: ti = 23, ar = 14, es = 4.

Tigrigna (ti, layer 23)			Arabic (ar, layer 14)			Spanish (es)	
Profession	Gloss	Proj	Profession	Gloss	Proj	Profession	Proj
ዳኛ	judge	+7.13	عالِم	scientist	+6.38	profesor	+2.79
መምህር	teacher	+6.89	معلم	teacher	+6.33	arquitecto	+2.44
ፖሊስ	police officer	+6.11	قاضی	judge	+5.94	agricultor	+2.22
ኮኮ	cook/chef	+6.03	مدیر	manager	+5.89	piloto	+2.15
ኣረስታይ	farmer	+5.72	مهندس	engineer	+5.67	cocinero	+2.13

4.5 Estimator Agreement (RQ3)

Table 4 reports Spearman rank correlation (Eq. 7) between the projection and cosine-difference estimators across all layers for each model.

Table 4: Layer-wise Spearman ρ between proj and cosdiff estimators, per model and language. Min/Max computed over all valid layers (layer 12 excluded for XLM-V and mDeBERTa where representational collapse occurs; see Discussion).

Model	Tigrigna (ti)		Arabic (ar)		Spanish (es)	
	Min	Max	Min	Max	Min	Max
XLM-R-base	0.992	1.000	0.964	1.000	0.973	1.000
XLM-R-large	0.993	1.000	0.981	1.000	0.983	1.000
XLM-V-base	0.970	0.999	0.978	0.999	0.990	1.000
mDeBERTa-v3-base	0.974	0.997	0.991	0.999	0.971	0.999

For all four models and all three languages, Spearman $\rho \geq 0.964$ at every valid layer (addressing RQ3). The two estimators, one magnitude-sensitive (Eq. 4) and one angle-only (Eq. 5), agree on the ordinal ranking of professions despite their different geometric properties, confirming that the observed bias patterns are not artifacts of estimator choice.

4.6 Architectural Reversal: mDeBERTa and Disentangled Attention

The most striking result is a complete inversion of bias direction under mDeBERTa-v3-base. Whereas the XLM-R family produces 25–26 male-leaning professions per model in Tigrigna, mDeBERTa produces only 2 male-leaning professions and 28 female-leaning ones (addressing RQ4).

Table 5 lists the professions that remain male-leaning under mDeBERTa, averaged across all valid layers.

Table 5: Male-leaning professions under mDeBERTa-v3-base (Tigrigna and Arabic), mean projection across all valid layers (layer 12 excluded). All remaining professions are female-leaning.

Language	Profession	Gloss	Mean Proj
Tigrigna	ካረከታይ	farmer	+0.20
Tigrigna	ፕሮፌሰር	professor	+0.15
Arabic	عالِم	scientist	+0.83
Arabic	مدير	manager	+0.53
Arabic	معلم	teacher	+0.38
Arabic	مهندس	engineer	+0.34

Spanish shows a weaker reversal: 11 of 24 professions remain male-leaning (e.g. *piloto* +0.64, *arquitecto* +0.59), attributable to grammatically gendered profession pairs (e.g. *abogado/abogada*) that partially dilute the directional signal.

The layer-12 collapse is especially pronounced for mDeBERTa: individual professions reach values as extreme as -17.02 (ካረከታይ) and -16.58 (ተከኒሻን), far exceeding the XLM-R-large layer-24 collapse (mean -0.51). This suggests that the final hidden state in mDeBERTa undergoes a qualitatively different representational transition than in the XLM-R family.

We attribute the directional reversal to Disentangled Attention (He et al., 2021), which encodes content and position information in separate attention matrices. When content-based and position-based representations are disentangled, the gender direction $\mathbf{g}_\ell$ (constructed from anchor centroids, Eq. 3) may align orthogonally or even inversely with profession subspace geometry relative to standard multi-head attention. The result is that profession vectors cluster on the female side of the gender axis, not because mDeBERTa associates professions with femininity in any direct cultural sense, but because its internal geometry is organised differently.

4.7 Vocabulary Size and Corpus Alignment: XLM-V

XLM-V-base was designed to overcome the vocabulary bottleneck in XLM-R by expanding the SentencePiece vocabulary from 250K to 1M tokens (Liang et al., 2023). For most low-resource languages this improves tokenisation fidelity. However, for Tigrigna we observe a paradoxical misalignment: two profession terms, *ጠበቅ* (lawyer) and *ኮቻይ* (brick-layer), are tokenised differently by the 1M vocabulary than by the 250K vocabulary, producing subword sequences absent from the evaluation corpus. Both terms receive 0 corpus hits under XLM-V, and are excluded from the bias calculation.

This illustrates that larger vocabulary coverage does not guarantee better corpus alignment: if the corpus was generated or tokenised with a different vocabulary, enlarging the model vocabulary introduces new mismatches. The XLM-V results (28 professions, peak +2.63) are otherwise consistent with the XLM-R family, confirming that when tokenisation succeeds, bias patterns are robust to vocabulary size.

5 Discussion

5.1 Cross-Lingual Patterns

The Tigrigna results demonstrate that multilingual transformers encode gender associations even for languages absent from their dedicated pretraining data. This is consistent with Lauscher et al. (2020), who show that properties from high-resource languages transfer implicitly to low-resource ones via shared subword representations. Because Tigrigna tokens are processed via SentencePiece subword pieces that partially overlap with other Semitic or phonologically similar languages, the models inherit geometric associations from those languages’ training signal.

5.2 Cross-Lingual Differences

The strength of the bias differs in the three languages. Arabic has the strongest peak in all the models. For XLM-R-large, the peak is at +4.54 in Arabic, followed by +3.46 in Tigrigna, and +1.16 in Spanish. Arabic also has the strongest male sweep in the XLM-R-large model, where all 23 Arabic professions have a male-mean, and not a single profession has a female-mean. This is likely because Arabic and Tigrigna share more vocabulary at the subword level, not only with each other but also with the Semitic language family, which is well-represented in the training data.

Finally, Spanish has the widest range in male-to-female ratios in all the models. This is because the list of Spanish professions has grammatical gender, and the pairings of male and female occupations, such as *abogado* and *abogada*, have the layer mean being pulled in opposite directions, leading to a weaker signal overall. This is not because the bias is weaker in the Spanish model but because the language has grammatical gender.

Finally, mDeBERTa has the same pattern, strongest in Arabic at −3.04, followed by Tigrigna at −1.84, and the weakest in Spanish at −0.62.

5.3 Layer-Wise Interpretation

The consistent pattern across the XLM-R family, building from layer 0 through mid-layers, peaking at layer 7–11 (base) or layer 23 (large), then collapsing, aligns with the established view that lower layers encode syntactic and morphological features while upper layers encode semantic relationships (Tenney et al., 2019; Rogers et al., 2020). Gender associations, as social-semantic properties, are expected to peak in the layers where semantic organisation is most pronounced.

5.4 Architecture Determines Bias Direction (RQ4)

One interesting finding of this paper is the finding of the reversal in mDeBERTa’s results. mBERT, XLM-R-base, XLM-R-large, and XLM-V-base show male-bias in their association for the same set of Tigrigna

professions, as trained on broadly comparable multilingual corpora. On the other hand, mDeBERTa-v3-base, as trained on comparable multilingual corpora, shows a female-bias for 28 of the same 30 professions.

One of the main differences in the architecture of mDeBERTa and XLM-R is in the attention mechanism, where Disentangled Attention is used in mDeBERTa, and multi-head attention is used in XLM-R. This finding demonstrates that *architecture dominates bias direction* in the low-resource cross-lingual setting, even when the data has a comparable pretraining setting. The practical implication of this is that mitigation of bias, as performed by data curation, may not be as useful as it appears, as the attention mechanism has a larger impact on the geometric association.

5.5 Metric Robustness

The agreement between the projection estimator (Eq. 4) and the cosine-difference estimator (Eq. 5) at $\rho \geq 0.964$ across all valid models is a methodological positive. The two estimators capture different geometric properties: one is magnitude-sensitive, the other is angle-only, yet they rank professions in nearly identical order at every layer. This confirms that the bias ranking is a robust geometric feature of the model’s representation space, not an artifact of the particular estimator chosen.

5.6 Limitations

- **Corpus construction:** The Tigrigna corpus is template-based, controlling for surface-level context but limiting ecological validity. Bias measurements may differ on naturally occurring text.
- **Tokenisation and morphology:** Tigrigna is morphologically rich; profession terms may be systematically fragmented across models, affecting the subword mean-pooling step.
- **Anchor coverage:** The 21-word anchor set (11 male, 10 female) is necessarily limited. Different anchor choices may shift the gender direction and alter profession rankings.
- **Near-static representations:** Tigrigna tokens produce highly similar representations across contexts due to template-based construction. The bias signal reflects intrinsic token geometry rather than contextually modulated gender inference.
- **Causal opacity:** We cannot directly observe the mechanism by which Disentangled Attention reverses bias direction; the attribution to attention architecture is inferential.

6 Conclusion

This study undertakes a systematic and layer-wise analysis of gender bias in five multilingual transformer architectures for three languages: Tigrigna, a low-resource Ge’ez-script language, Arabic, and Spanish, by means of a unified geometric framework in the context of contextual-to-static reductions (Bommasani et al., 2020) and projection-based bias estimation (Bolukbasi et al., 2016). Focusing on Tigrigna as the main case study, we arrive at four main conclusions.

First, three of four models encode measurable male-leaning gender associations for Tigrigna professions despite the language’s absence from dedicated pretraining data, confirming cross-lingual transfer of social associations (RQ1). Second, bias accumulates progressively across transformer depth, peaking at mid-to-late layers and collapsing at the final layer, a pattern consistent with the semantic organisation of transformer representations (RQ2). Third, two independent geometric estimators agree at Spearman $\rho \geq 0.964$ across all valid models, confirming methodological robustness (RQ3). Fourth, and most importantly, mDeBERTa-v3-base’s Disentangled Attention mechanism completely inverts the bias direction relative to the XLM-R family, producing female-leaning associations for 28 of 30 professions, strong evidence that architectural choice can override data-driven gender associations (RQ4).

These results have the following practical consequences: the coverage of SentencePiece vocabularies is a necessary condition for bias analysis in low-resource languages, bias audits should consider intermediate layers, not the final layer, and mitigation of bias in the pretraining data may not suffice, especially in the

presence of an independent influence of the attention mechanism on bias geometry.

7 Future Work

- Collect small-scale human judgments from Tigrigna speakers to validate profession-gender associations against cultural norms.
- Probe the Disentangled Attention mechanism mechanistically (attention-head attribution, representation similarity analysis) to understand why it reverses bias direction.
- Test whether post-hoc debiasing methods ([Bolukbasi et al., 2016](#)) applied at the intermediate peak layer reduce downstream bias in classification tasks.
- Evaluate additional low-resource languages with Ge'ez and other non-Latin scripts to assess the generalisability of the findings.

References

- Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, and Adam Kalai. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings. In *Advances in Neural Information Processing Systems*, volume 29, 2016.
- Rishi Bommasani, Kelly Davis, and Claire Cardie. Interpreting pretrained contextualized representations via reductions to static embeddings. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4758–4781. Association for Computational Linguistics, 2020.
- Aylin Caliskan, Joanna J. Bryson, and Arvind Narayanan. Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334):183–186, 2017.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Édouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451. Association for Computational Linguistics, 2020.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 4171–4186. Association for Computational Linguistics, 2019.
- Nikhil Garg, Londa Schiebinger, Dan Jurafsky, and James Zou. Word embeddings quantify 100 years of gender and ethnic stereotypes. *Proceedings of the National Academy of Sciences*, 115(16):E3635–E3644, 2018.
- Hila Gonen and Yoav Goldberg. Lipstick on a pig: Debiasing methods cover up systematic gender biases in word embeddings but do not remove them. In *Proceedings of the 2019 Workshop on Gender Bias in Natural Language Processing*. Association for Computational Linguistics, 2019.
- Wei Guo and Aylin Caliskan. Detecting emergent intersectional biases: Contextualized word embeddings contain a distribution of human-like biases. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, pages 122–133, 2021.
- Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. DeBERTa: Decoding-enhanced BERT with disentangled attention. In *International Conference on Learning Representations*, 2021.
- Pengcheng He, Jianfeng Gao, and Weizhu Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *International Conference on Learning Representations*, 2023.
- Anne Lauscher, Vinit Ravishankar, Ivan Vulić, and Goran Glavaš. From zero to hero: On the limitations of zero-shot language transfer with multilingual Transformers. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 4483–4499, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.363. URL <https://aclanthology.org/2020.emnlp-main.363/>.
- Davis Liang, Hila Xu, Punit Singh Shi, Srinivasan Iyer, Subhabrata Ghosh, Luke Zettlemoyer, and Yashar Mehdad. XLM-V: Overcoming the vocabulary bottleneck in multilingual masked language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13596–13610. Association for Computational Linguistics, 2023.
- Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. *Introduction to Information Retrieval*. Cambridge University Press, 2008.

Anna Rogers, Olga Kovaleva, and Anna Rumshisky. A primer in BERTology: What we know about how BERT works. *Transactions of the Association for Computational Linguistics*, 8:842–866, 2020.

Charles Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904.

Ian Tenney, Dipanjan Das, and Ellie Pavlick. BERT rediscovers the classical NLP pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4593–4601. Association for Computational Linguistics, 2019.

Jieyu Zhao, Tianlu Wang, Mark Yatskar, Ryan Cotterell, Vicente Ordonez, and Kai-Wei Chang. Gender bias in contextualized word embeddings. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 629–634, 2019.

Jieyu Zhao, Subhabrata Mukherjee, Saghar Hosseini, Kai-Wei Chang, and Ahmed Hassan Awadallah. Gender bias in multilingual embeddings and cross-lingual transfer. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2896–2907, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.260. URL <https://aclanthology.org/2020.acl-main.260/>.

A mBERT Degeneration (Negative Control)

mBERT’s WordPiece vocabulary does not cover Tigrigna characters; all tokens are mapped to [UNK]. This produces two degenerate clusters at every layer: 26 single-word professions share one identical projection value per layer (e.g., -3.614 at layer 0), and the 4 multi-word professions share a distinct but also uniform value (-0.677 at layer 0). No profession-level differentiation exists.

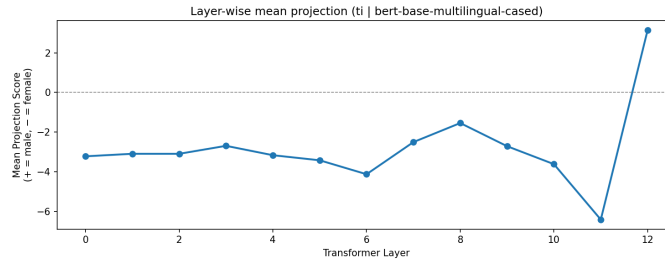


Figure 6: mBERT on Tigrigna: WordPiece [UNK] degeneration. Two flat clusters are visible; no profession-level signal.

mBERT therefore serves as a *negative control*, confirming that SentencePiece vocabulary coverage is a prerequisite for meaningful bias measurement in low-resource Ge’ez-script languages.

B Profession Inventory

B.1 Full Tigrigna Profession List (30 Terms)

- ሓኪም (doctor)
- ነርስ (nurse)
- መምህር (teacher)
- ፕሮፌሰር (professor)
- መሃንዲስ (engineer)
- ሳይንቲስት (scientist)
- ጠበቅ (lawyer)

- ዳኛ (judge)
- ሓላፊ (manager)
- ዳይሬክተር (director)
- ወተሃደር (soldier)
- ፖሊስ (police officer)
- ሓረስታይ (farmer)
- ነጋዴይ (merchant)
- መካኒክ (mechanic)
- ኤሌክትሪሻን (electrician)
- ፈላጃዊ (pilot)
- ነዳቻይ (brick-layer)
- አርኪቴክት (architect)
- አካውንታንት (accountant)
- ባንከር (banker)
- ስነ ጥበባዊ (artist)
- መዝቀኛ (musician)
- ተዋሳኢ (actor)
- ስራሕ ፈጣሪ (entrepreneur)
- ኮኮ (cook/chef)
- ስራሕተኛ ጽሬት (cleaning worker)
- ሓላዊ ጸጥታ (security guard)
- ተክኒሻን (technician)
- ፕሮግራመር (programmer)

B.2 Sample Gender Anchors (Tigrigna)

Male anchors (11): pronouns, kinship terms, and generic masculine markers in Tigrigna (e.g., ንሱ ‘he’, ሓወይ ‘my brother’, እቦ ‘father’, ወዲ ‘son/boy’).

Female anchors (10): corresponding feminine forms (e.g., ንሳ ‘she’, ሓፍተይ ‘my sister’, እደ ‘mother’, ንል ‘daughter/girl’).

Declaration of Academic Integrity

I,

Last name, first name: KEBEDE, DANIEL TESFAI

Student ID number: 1716694

hereby declare that I have written this thesis independently and have not used any sources or aids other than those specified. All direct or indirect content and ideas from external sources are acknowledged as references. I confirm that I have not previously submitted this thesis in part or whole for any other academic credit nor has the work been submitted to any other examination office and has not been published.

I confirm that I have only used text or other content-generating aids based on generative artificial intelligence (GenAI) in the manner permitted in writing by the examiner and that I have explicitly marked GenAI-generated passages.

When using GenAI tools, I have indicated these with their product name and listed them in full in an appendix.

I am fully responsible for any GenAI-generated passages and content I have used in my work.

Thesis title: Layer-Wise Gender Bias in Multilingual Foundation Models: A Five-Architecture Cross-Lingual Analysis

Programme: MSC, NATURAL LANGUAGE PROCESSING

Examiner: RAGHVI BALONI

This is an achievement for a

☐ Bachelor's thesis

☐ Master's thesis

☒ other examination achievement for module (term paper etc.) Term paper, Trends in Machine Learning, (Winter Semester 2025/26)

TRIER, 31-03-2026

Place, Date

Signature

